Block practice · Networks, media, protocols and the internet

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Network types: LAN, WAN, PAN and why we network

Network protocols: what they are and which one does what

Network topologies: bus, ring, star and mesh

The internet, the web, DNS and the parts of a URL

Wired or wireless? Media, pros and cons, and network speed units

The TCP/IP model: four layers and what each does

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Practice workbook · Network types: LAN, WAN, PAN and why we network

Reasons for networking computers; naming or contrasting LAN, WAN and PAN from a described scenario; client-server versus peer-to-peer usage models.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-01

1. Question 1 · part a · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

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1. Question 1 · part a · 2 marks · Working and answer

Working and answer

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2. Question 3 · part a.i · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.i only.

3 Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

.....

.....

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

.....

.....

(b) Identify the measurement used for network data speeds.

(1)

☒ **A** Megabits per second

☒ **B** Megapixels per second

☒ **C** Mebicycles per second

☒ **D** Mebihertz per second

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2. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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3. Question 3 · part a.ii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.ii only.

3 Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

.....

.....

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

.....

.....

(b) Identify the measurement used for network data speeds.

(1)

☒ **A** Megabits per second

☒ **B** Megapixels per second

☒ **C** Mebicycles per second

☒ **D** Mebihertz per second

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3. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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4. Question 1 · part a · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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2

4. Question 1 · part a · 2 marks · Working and answer

Working and answer

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5. Question 2 · part a · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

P 6 6 4 8 9 A 0 5 2 0

5. Question 2 · part a · 1 mark · Working and answer

Working and answer

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6. Question 3 · part a · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

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3 Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

(2)

Network type	Tick (✓)	Usage model	Tick (✓)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

	Layer name	Function
1		
2		

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6. Question 3 · part a · 2 marks · Working and answer

Working and answer

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7. Question 5 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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16

P 7 2 5 3 8 A 0 1 6 2 0

7. Question 5 · part b.i · 1 mark · Working and answer

Working and answer

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8. Question 5 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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8. Question 5 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

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(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

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(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

Description	RAM	ROM
Stores the boot up sequence		
The contents are lost when the laptop is shut down		

(Total for Question 5 = 12 marks)

8. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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9. Question 6 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

6 Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

Figure 3 shows a network topology.

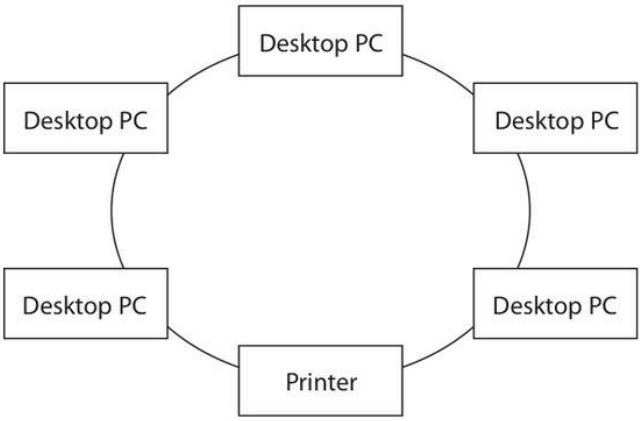


Figure 3

(i) Explain **one** benefit of this network topology. **(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet. **(2)**

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P 7 2 5 3 8 A 0 1 8 2 0

9. Question 6 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

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(b) Santiago works on his laptop whilst travelling by train.

There is a free Wi-Fi connection on the train, but Santiago doesn't use it.

He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

(1)

(ii) Explain **one** advantage for Santiago of using the network he has set up to connect to the internet, rather than the free Wi-Fi connection.

(2)

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

(1)

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

(1)

9. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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10. Question 1 · part a · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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10. Question 1 · part a · 2 marks · Working and answer

Working and answer

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Practice workbook · Network protocols: what they are and which one does what

Why protocols are needed and what named protocols do, including choosing and justifying an email protocol (IMAP/POP3/SMTP) and describing Ethernet.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-03

1. Question 3 · part c · 3 marks

2019 JUN · Paper 01

Worked-example retry · Attempt part c only.

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3 Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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1. Question 3 · part c · 3 marks

2019 JUN · Paper 01

Worked-example retry · Attempt part c only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	☒	☒	☒	☒
All devices are directly connected to all others.	☒	☒	☒	☒
Each node has a physical attachment to a routing device.	☒	☒	☒	☒
There are no data collisions, because packets travel in the same direction.	☒	☒	☒	☒

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P 6 1 8 8 0 A 0 8 2 4

1. Question 3 · part c · 3 marks · Working and answer

Working and answer

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2. Question 1 · part f.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.i only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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2

P 6 1 8 8 4 R R A 0 2 2 0

2. Question 1 · part f.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.i only.

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(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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2. Question 1 · part f.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.i only.

(e) Network layouts can be based upon a number of different network topologies.

Figure 1 shows a network topology.

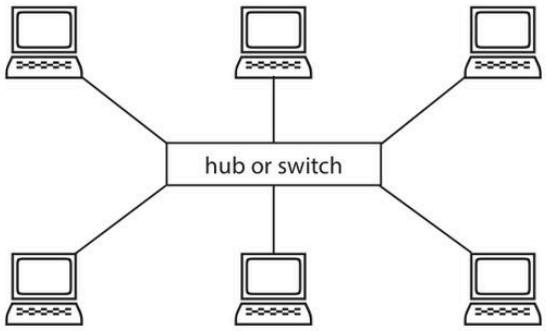


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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P 6 1 8 8 4 R R A 0 4 2 0

2. Question 1 · part f.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.i only.

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(f) The network uses TCP/IP.

Figure 2 shows the TCP/IP protocol stack.

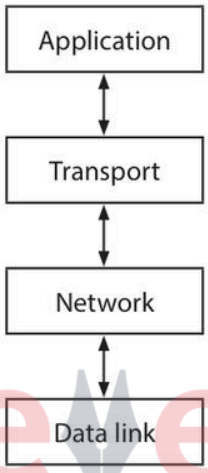


Figure 2

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(i) One use of the application layer is to send and receive emails.

State **two** email protocols.

(2)

1 .....

2 .....

(ii) Upeka sends an email to Peter. The transport layer in Upeka's computer splits the email into data packets.

State **two** tasks performed by the transport layer in Peter's computer when the packets arrive from the network layer.

(2)

1 .....

.....

2 .....

.....

(Total for Question 1 = 17 marks)

2. Question 1 · part f.i · 2 marks · Working and answer

Working and answer

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3. Question 2 · part b · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b only.

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

3. Question 2 · part b · 1 mark · Working and answer

Working and answer

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4. Question 4 · part c.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c.ii only.

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4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

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(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

4. Question 4 · part c.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c.ii only.

(ii) Some people confuse Ethernet® and Wi-Fi.

Describe what is meant by the term **Ethernet®**.

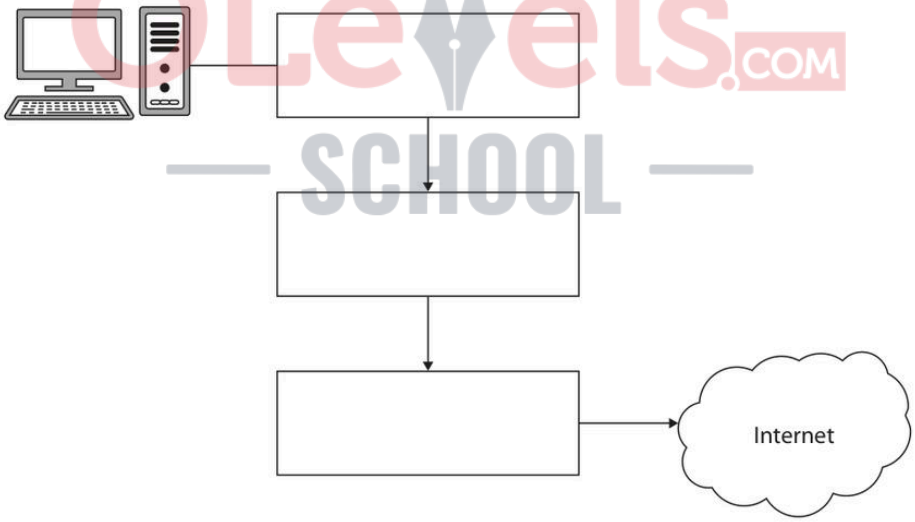
(2)

(d) A single physical box connects a desktop computer to the Internet.

The box incorporates three different components.

Complete the diagram to show the names of the components in the correct order.

(3)



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P 7 2 9 3 7 A 0 1 2 2 4

4. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

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5. Question 1 · part c · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part c only.

1 Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

(4)

Statement	DNS	PAN	WAN	WWW
A network of networks **not** limited to a single location	☒	☒	☒	☒
A network that connects devices within the range of a single user	☒	☒	☒	☒
A set of hypertext documents accessed through the HTTP protocol	☒	☒	☒	☒
A system that maps domain names to IP addresses	☒	☒	☒	☒

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

(2)

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

(1)

- ☒ **A** HTTP
- ☒ **B** IMAP
- ☒ **C** POP3
- ☒ **D** SMTP

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2

P 7 8 6 3 7 A 0 2 2 0

5. Question 1 · part c · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part c only.

(d) A library sets up a network using a star topology.

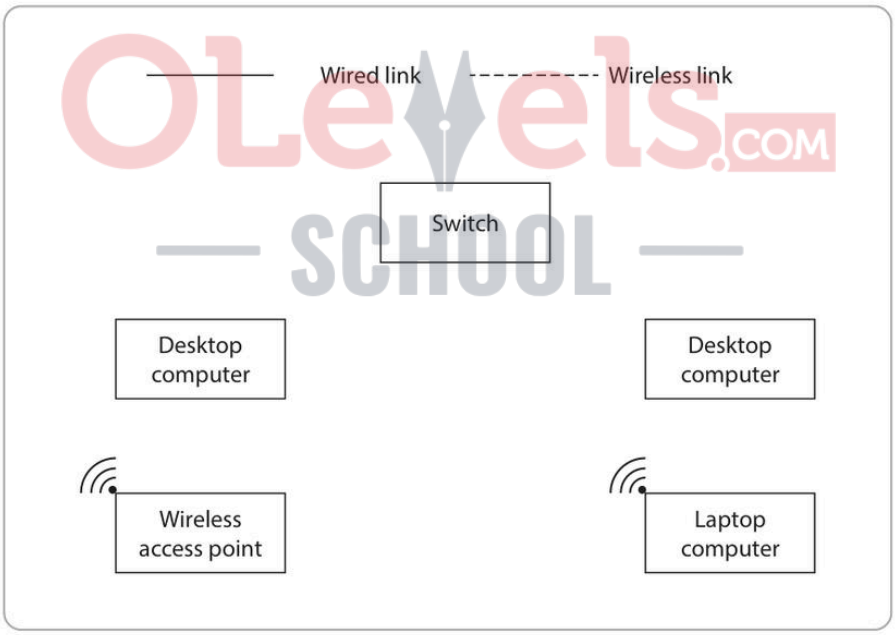
The library is connecting:

- two desktop computers with wired network interface cards
- a laptop with a wireless network interface card
- a switch
- a wireless access point.

Complete the diagram to show how the devices are connected.

Use a solid line to indicate a wired link and a dashed line to indicate a wireless link.

(3)



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5. Question 1 · part c · 1 mark · Working and answer

Working and answer

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Practice workbook · Network topologies: bus, ring, star and mesh

Recognising a topology from a diagram or description and stating/explaining its benefits, drawbacks and fault tolerance.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-05

1. Question 1 · part c.iii · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c.iii only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

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P 7 5 7 3 6 A 0 2 2 8

1. Question 1 · part c.iii · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c.iii only.

DO NOT WRITE IN THIS AREA

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(c) **Figure 1** shows a network topology.

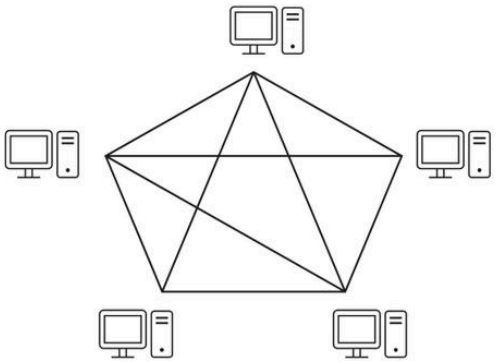


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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1. Question 1 · part c.iii · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c.iii only.

(d) Identify which layer of the 4-layer TCP/IP model is responsible for transferring packets across a network.

(1)

- ☒ **A** Application layer
- ☒ **B** Transport layer
- ☒ **C** Network layer
- ☒ **D** Data link layer

(e) Here is the address for a web page.

https://www.pearson.com/computing/specification.html

(i) State the protocol being used.

(1)

(ii) Give the name of the document being retrieved.

(1)

(Total for Question 1 = 10 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN

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Working and answer

2. Question 3 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

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3 Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

.....

.....

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

.....

.....

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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2. Question 3 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	☒	☒	☒	☒
All devices are directly connected to all others.	☒	☒	☒	☒
Each node has a physical attachment to a routing device.	☒	☒	☒	☒
There are no data collisions, because packets travel in the same direction.	☒	☒	☒	☒

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8

P 6 1 8 8 0 A 0 8 2 4

2. Question 3 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

(e) Information sent across networks is represented in bit patterns.

(i) The bit pattern 1101 0001 uses sign and magnitude representation.

Convert this bit pattern to a denary number.

(2)

(ii) Convert the denary number 75 to 8-bit binary.

(2)

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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2. Question 3 · part d · 4 marks · Working and answer

Working and answer

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3. Question 1 · part e.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.i only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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2

P 6 1 8 8 4 R R A 0 2 2 0

3. Question 1 · part e.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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3. Question 1 · part e.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.i only.

(e) Network layouts can be based upon a number of different network topologies.

Figure 1 shows a network topology.

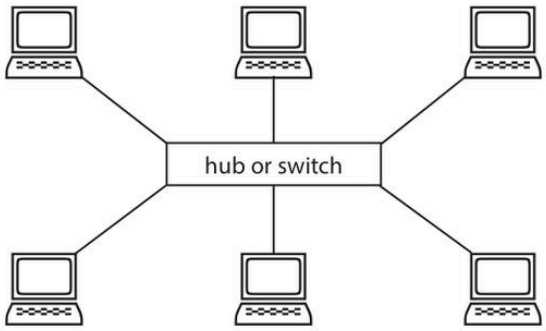


Figure 1

(i) State the name of this network topology. (1)

(ii) State **one** benefit of using this network topology. (1)

(iii) State **one** drawback of using this network topology. (1)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
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3. Question 1 · part e.i · 1 mark · Working and answer

Working and answer

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4. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.ii only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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2

P 6 1 8 8 4 R R A 0 2 2 0

4. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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4. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.ii only.

(e) Network layouts can be based upon a number of different network topologies.

Figure 1 shows a network topology.

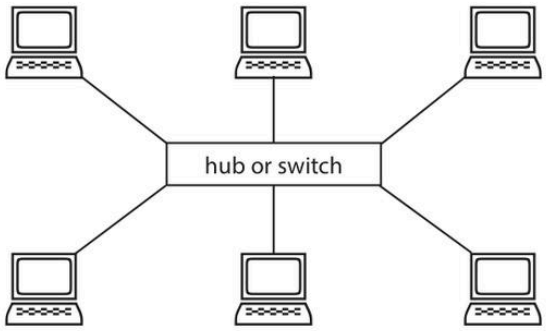


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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DO NOT WRITE IN THIS AREA

P 6 1 8 8 4 R R A 0 4 2 0

4. Question 1 · part e.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part e.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.iii only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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2

P 6 1 8 8 4 R R A 0 2 2 0

5. Question 1 · part e.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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5. Question 1 · part e.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.iii only.

(e) Network layouts can be based upon a number of different network topologies.

Figure 1 shows a network topology.

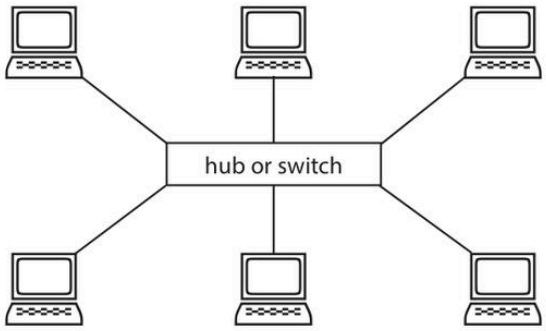


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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DO NOT WRITE IN THIS AREA

P 6 1 8 8 4 R R A 0 4 2 0

5. Question 1 · part e.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.iii only.

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(f) The network uses TCP/IP.

Figure 2 shows the TCP/IP protocol stack.

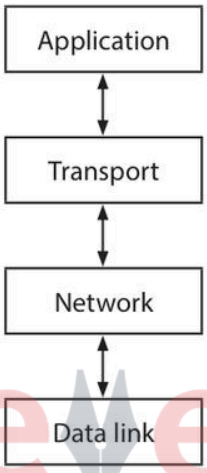


Figure 2

(i) One use of the application layer is to send and receive emails.

State **two** email protocols.

(2)

1 .....

2 .....

(ii) Upeka sends an email to Peter. The transport layer in Upeka's computer splits the email into data packets.

State **two** tasks performed by the transport layer in Peter's computer when the packets arrive from the network layer.

(2)

1 .....

.....

2 .....

.....

(Total for Question 1 = 17 marks)

5. Question 1 · part e.iii · 1 mark · Working and answer

Working and answer

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6. Question 2 · part e.i · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.i only.

DO NOT WRITE IN THIS AREA

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

(b) State the purpose of network protocols.

(1)

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

(ii) Give **one** task carried out by the network layer.

(1)

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

OLevels.COM SCHOOL

P 6 6 4 8 9 A 0 5 2 0

6. Question 2 · part e.i · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.i only.

(e) **Figure 2** shows a network topology.

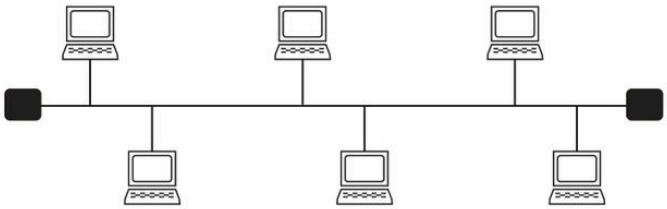


Figure 2

(i) State the name of this network topology.

(1)

.....

(ii) State **one** benefit of using this network topology.

(1)

.....

.....

(iii) State **one** drawback of using this network topology.

(1)

.....

.....

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1 .....

.....

2 .....

.....

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DO NOT WRITE IN THIS AREA

P 6 6 4 8 9 A 0 6 2 0

6. Question 2 · part e.i · 1 mark · Working and answer

Working and answer

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7. Question 2 · part e.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

(b) State the purpose of network protocols.

(1)

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

(ii) Give **one** task carried out by the network layer.

(1)

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

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7. Question 2 · part e.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.ii only.

(e) **Figure 2** shows a network topology.

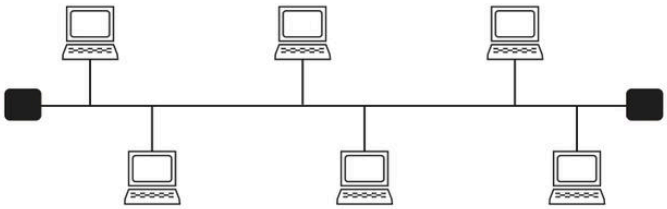


Figure 2

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1 .....

2 .....

DO NOT WRITE IN THIS AREA

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6

P 6 6 4 8 9 A 0 6 2 0

7. Question 2 · part e.ii · 1 mark · Working and answer

Working and answer

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8. Question 2 · part e.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.iii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

P 6 6 4 8 9 A 0 5 2 0

8. Question 2 · part e.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.iii only.

(e) **Figure 2** shows a network topology.

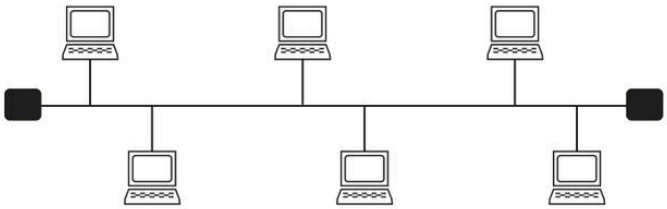


Figure 2

(i) State the name of this network topology.

(1)

.....

(ii) State **one** benefit of using this network topology.

(1)

.....

.....

(iii) State **one** drawback of using this network topology.

(1)

.....

.....

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(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1 .....

.....

2 .....

.....

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8. Question 2 · part e.iii · 1 mark · Working and answer

Working and answer

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9. Question 6 · part a.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.i only.

6 Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

Figure 3 shows a network topology.

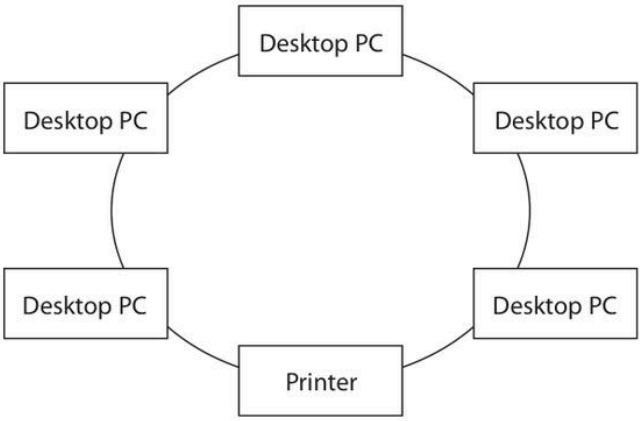


Figure 3

(i) Explain **one** benefit of this network topology. **(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet. **(2)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 8 2 0

9. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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10. Question 6 · part a.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

6 Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

Figure 3 shows a network topology.

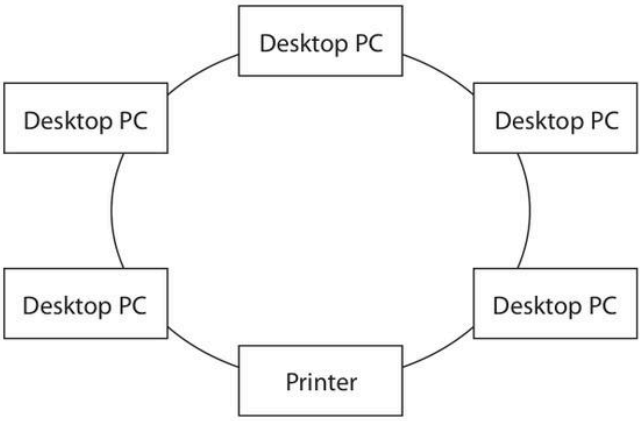


Figure 3

(i) Explain **one** benefit of this network topology.

(2)

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10. Question 6 · part a.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

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(b) Santiago works on his laptop whilst travelling by train.

There is a free Wi-Fi connection on the train, but Santiago doesn't use it.

He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

(1)

(ii) Explain **one** advantage for Santiago of using the network he has set up to connect to the internet, rather than the free Wi-Fi connection.

(2)

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

(1)

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

(1)

10. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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11. Question 1 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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11. Question 1 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(c) **Figure 1** shows a network topology.

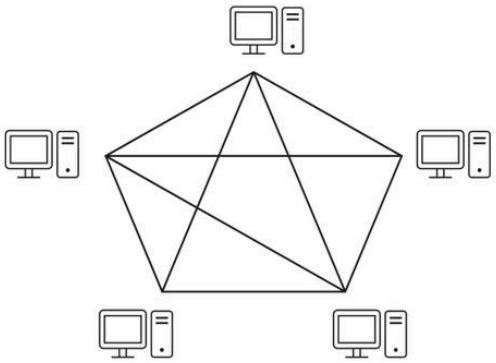


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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11. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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12. Question 1 · part c.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.ii only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

12. Question 1 · part c.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) **Figure 1** shows a network topology.

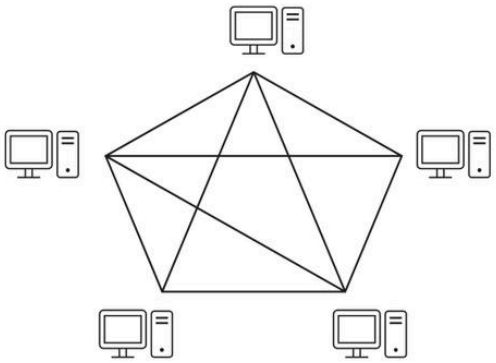


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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12. Question 1 · part c.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · The internet, the web, DNS and the parts of a URL

What the internet and WWW are, how DNS resolves names, and breaking a URL into protocol, domain, folder and file or classifying URL/domain/IP items.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-11

1. Question 3 · part b.iv · 4 marks

2022 JUN · Paper 01

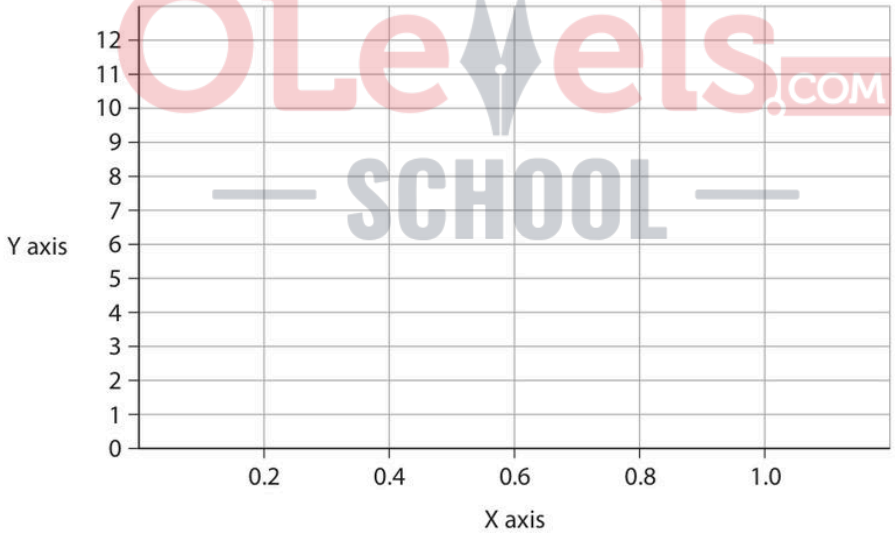
Worked-example retry · Attempt part b.iv only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

P 7 2 5 3 8 A 0 8 2 0

1. Question 3 · part b.iv · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part b.iv only.

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DO NOT WRITE IN THIS AREA

(b) Alyssa uploads music files to her cloud storage.

(i) She compresses the files before she uploads them using a lossless algorithm.

Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

(1)

(ii) Explain **one** benefit to Alyssa of storing her music files in the cloud.

(2)

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(iii) Give **one** possible security issue associated with storing music files in the cloud.

(1)

(iv) One of Alyssa's music files is stored at https://www.cloudisfab.com/re12/ru2.mp3

Complete the table by adding a description of each URL component.

(4)

URL component	Description
https	
www.cloudisfab.com	
re12	
ru2.mp3	

(Total for Question 3 = 13 marks)

1. Question 3 · part b.iv · 4 marks · Working and answer

Working and answer

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2. Question 2 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

2 Zafer and Robert work for a company that makes washing machines.

(a) Zafer writes user manuals for the washing machines.

He stores these documents in the cloud.

Zafer and the cloud storage provider share responsibility for data security.

State **one** area of responsibility for each of them.

(2)

Zafer

.....

.....

Cloud storage provider

.....

.....

(b) Robert programs the robots that make the parts for the washing machines.

Zafer writes the user manuals.

Robert and Zafer have different levels of access to folders and files on the company network.

The different levels of access are Read, Write, Execute and None.

Complete the table to show the levels of access that each has.

(2)

	Robert	Zafer
Washing machine design drawings		
A folder of manuals for new machines		

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P 6 1 8 8 0 A 0 4 2 4

2. Question 2 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

(c) Zafer uses a browser to view pages on Pearson's website by typing a uniform resource locator into a browser. This is shown in **Figure 1**.



Figure 1

Pearson's machine	Zafer's machine
MAC address: 03-57-00-EC-4B-30	MAC address: 00-15-00-BC-9A-90
IPv4 address: 2.20.38.113	IPv4 address: 192.168.1.78
IPv4 Subnet Mask: 255.255.0.0	IPv4 Subnet Mask: 255.255.255.0
Website domain: www.pearson.com	Web browser
Main web page: https://www.pearson.com/uk/	Uniform resource locator: www.pearson.com

A domain name server is used in this process.

Identify the input to and the output from the domain name server.

(2)

Input

Output

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2. Question 2 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

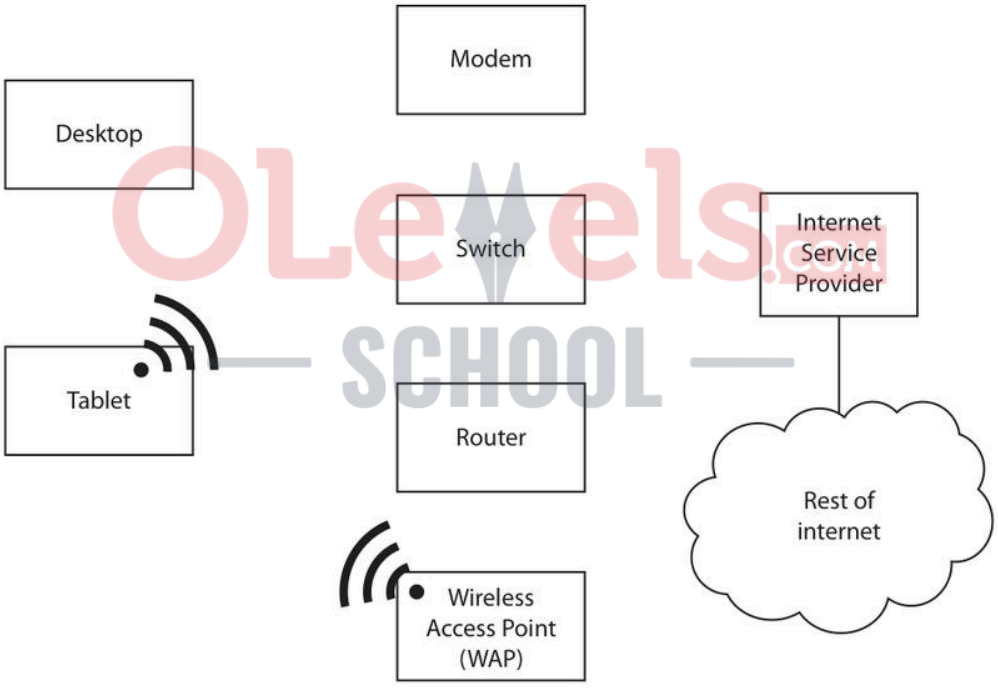
(d) Zafer can use a desktop or a tablet computer to connect to the internet.

Only the tablet has wireless capabilities.

The diagram shows the components used to connect to the internet.

Complete the diagram to show how the desktop and the tablet are connected to the internet.

(6)



(Total for Question 2 = 12 marks)

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2. Question 2 · part c · 2 marks · Working and answer

Working and answer

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3. Question 1 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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2

P 6 1 8 8 4 R R A 0 2 2 0

3. Question 1 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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3. Question 1 · part c · 2 marks · Working and answer

Working and answer

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4. Question 2 · part d · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

4. Question 2 · part d · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

(e) **Figure 2** shows a network topology.

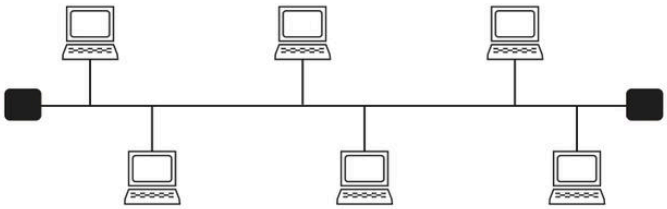


Figure 2

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1

2

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6

P 6 6 4 8 9 A 0 6 2 0

4. Question 2 · part d · 3 marks · Working and answer

Working and answer

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5. Question 1 · part e.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part e.i only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

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5. Question 1 · part e.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part e.i only.

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DO NOT WRITE IN THIS AREA

(c) **Figure 1** shows a network topology.

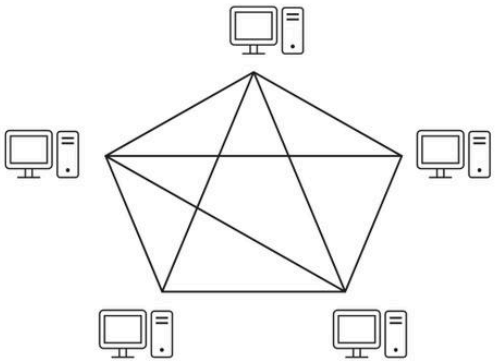


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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5. Question 1 · part e.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part e.i only.

(d) Identify which layer of the 4-layer TCP/IP model is responsible for transferring packets across a network.

(1)

- ☒ **A** Application layer
- ☒ **B** Transport layer
- ☒ **C** Network layer
- ☒ **D** Data link layer

(e) Here is the address for a web page.

https://www.pearson.com/computing/specification.html

(i) State the protocol being used.

(1)

.....

(ii) Give the name of the document being retrieved.

(1)

.....

(Total for Question 1 = 10 marks)

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Working and answer

6. Question 1 · part e.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part e.ii only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

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6. Question 1 · part e.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part e.ii only.

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(c) **Figure 1** shows a network topology.

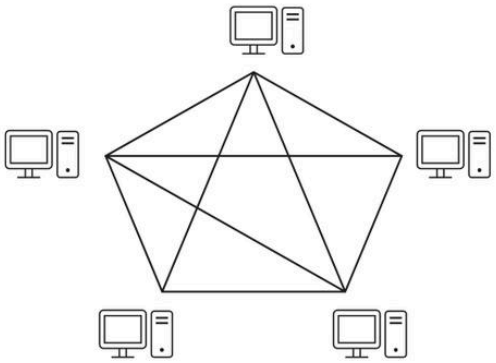


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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6. Question 1 · part e.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part e.ii only.

(d) Identify which layer of the 4-layer TCP/IP model is responsible for transferring packets across a network.

(1)

- ☒ **A** Application layer
- ☒ **B** Transport layer
- ☒ **C** Network layer
- ☒ **D** Data link layer

(e) Here is the address for a web page.

https://www.pearson.com/computing/specification.html

(i) State the protocol being used.

(1)

(ii) Give the name of the document being retrieved.

(1)

(Total for Question 1 = 10 marks)

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Working and answer

7. Question 1 · part a · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

1 Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

(4)

Statement	DNS	PAN	WAN	WWW
A network of networks **not** limited to a single location	☒	☒	☒	☒
A network that connects devices within the range of a single user	☒	☒	☒	☒
A set of hypertext documents accessed through the HTTP protocol	☒	☒	☒	☒
A system that maps domain names to IP addresses	☒	☒	☒	☒

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

(2)

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

(1)

☒ **A** HTTP

☒ **B** IMAP

☒ **C** POP3

☒ **D** SMTP

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7. Question 1 · part a · 4 marks · Working and answer

Working and answer

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Practice workbook · Wired or wireless? Media, pros and cons, and network speed units

Advantages and drawbacks of wired versus wireless links, the transmission media used, Wi-Fi frequency bands, units of network speed (bits per second), and choosing a private hotspot over public Wi-Fi.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-02

1. Question 4 · part c.i · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

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4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

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(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

.....

.....

.....

.....

1. Question 4 · part c.i · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

(ii) Some people confuse Ethernet® and Wi-Fi.

Describe what is meant by the term **Ethernet®**.

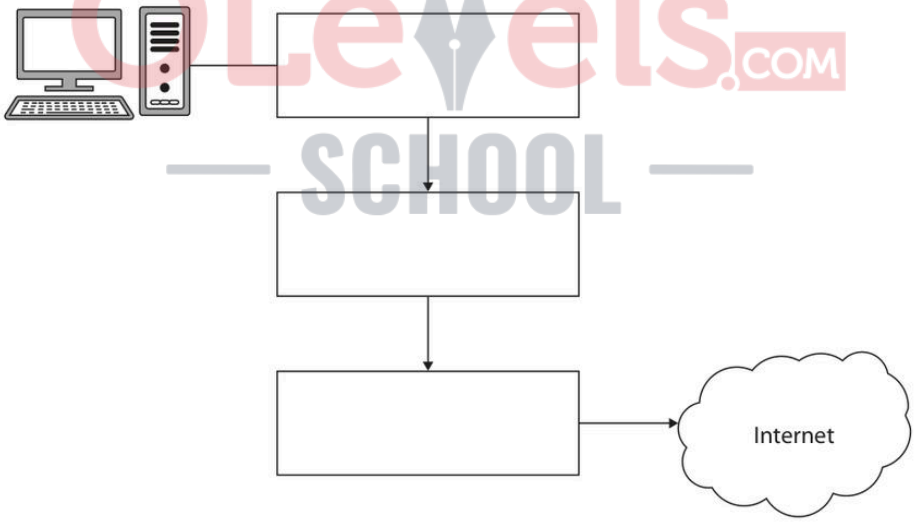
(2)

(d) A single physical box connects a desktop computer to the Internet.

The box incorporates three different components.

Complete the diagram to show the names of the components in the correct order.

(3)



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P 7 2 9 3 7 A 0 1 2 2 4

1. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

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2. Question 3 · part b · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part b only.

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3 Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

.....

.....

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

.....

.....

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

OLevels.COM SCHOOL

2. Question 3 · part b · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part b only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	☒	☒	☒	☒
All devices are directly connected to all others.	☒	☒	☒	☒
Each node has a physical attachment to a routing device.	☒	☒	☒	☒
There are no data collisions, because packets travel in the same direction.	☒	☒	☒	☒

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2. Question 3 · part b · 1 mark · Working and answer

Working and answer

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3. Question 1 · part b.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b.i only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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3. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 1 · part b.ii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b.ii only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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4. Question 1 · part b.ii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

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(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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4. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part c · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c only.

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3 Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

(2)

Network type	Tick (✓)	Usage model	Tick (✓)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

	Layer name	Function
1		
2		

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5. Question 3 · part c · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

- ☒ **A** 2.4 GHz
- ☒ **B** 3 KHz
- ☒ **C** 4.1 GHz
- ☒ **D** 5 KHz

(d) Carlo's smartphone can use 3G, 4G and 5G mobile communication standards.

(i) Give **one** advantage of using a higher frequency band for mobile communications.

(1)

(ii) Explain **one** benefit to Carlo of using 5G rather than 3G.

(2)

(e) Carlo wants to reduce the environmental impact of his smartphone.

(i) Give **one** way Carlo could reduce energy consumption.

(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

(Total for Question 3 = 13 marks)

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5. Question 3 · part c · 1 mark · Working and answer

Working and answer

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6. Question 6 · part b.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

6 Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

Figure 3 shows a network topology.

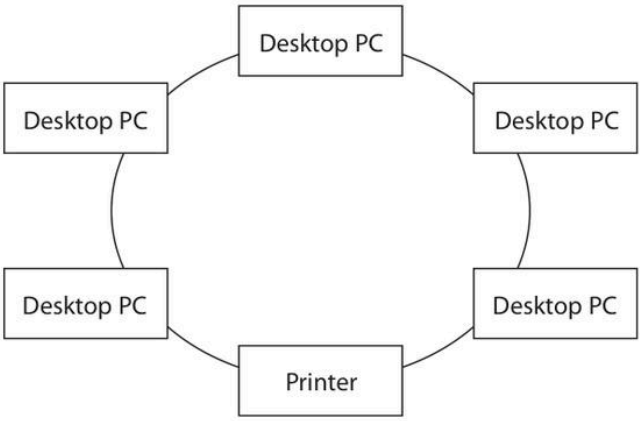


Figure 3

(i) Explain **one** benefit of this network topology. **(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet. **(2)**

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P 7 2 5 3 8 A 0 1 8 2 0

6. Question 6 · part b.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

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DO NOT WRITE IN THIS AREA

(b) Santiago works on his laptop whilst travelling by train.

There is a free Wi-Fi connection on the train, but Santiago doesn't use it.

He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

(1)

(ii) Explain **one** advantage for Santiago of using the network he has set up to connect to the internet, rather than the free Wi-Fi connection.

(2)

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

(1)

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

(1)

6. Question 6 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 4 · part a · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part a only.

4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

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7. Question 4 · part a · 1 mark · Working and answer

Working and answer

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Practice workbook · The TCP/IP model: four layers and what each does

Naming the four layers (application, transport, network, data link) and stating what each does to packets on sending and receiving.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-04

1. Question 3 · part b · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part b only.

DO NOT WRITE IN THIS AREA

3 Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

(2)

Network type	Tick (✓)	Usage model	Tick (✓)
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Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

DO NOT WRITE IN THIS AREA

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

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	Layer name	Function
1		
2		

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1. Question 3 · part b · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part b only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

- ☒ **A** 2.4 GHz
- ☒ **B** 3 KHz
- ☒ **C** 4.1 GHz
- ☒ **D** 5 KHz

(d) Carlo's smartphone can use 3G, 4G and 5G mobile communication standards.

(i) Give **one** advantage of using a higher frequency band for mobile communications.

(1)

(ii) Explain **one** benefit to Carlo of using 5G rather than 3G.

(2)

(e) Carlo wants to reduce the environmental impact of his smartphone.

(i) Give **one** way Carlo could reduce energy consumption.

(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

(Total for Question 3 = 13 marks)

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1. Question 3 · part b · 4 marks · Working and answer

Working and answer

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2. Question 1 · part f.ii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.ii only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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2

P 6 1 8 8 4 R R A 0 2 2 0

2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

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(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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2. Question 1 · part f.ii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.ii only.

(e) Network layouts can be based upon a number of different network topologies.

Figure 1 shows a network topology.

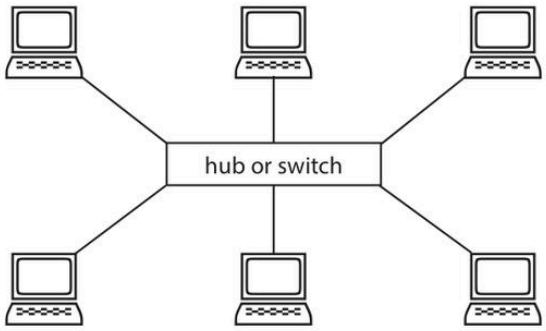


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

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(f) The network uses TCP/IP.

Figure 2 shows the TCP/IP protocol stack.

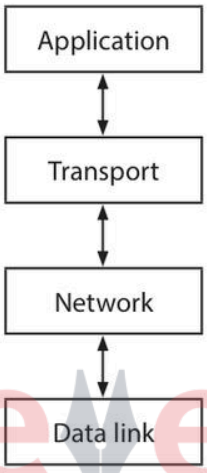


Figure 2

(i) One use of the application layer is to send and receive emails.

State **two** email protocols.

(2)

1 .....

2 .....

(ii) Upeka sends an email to Peter. The transport layer in Upeka's computer splits the email into data packets.

State **two** tasks performed by the transport layer in Peter's computer when the packets arrive from the network layer.

(2)

1 .....

.....

2 .....

.....

(Total for Question 1 = 17 marks)

2. Question 1 · part f.ii · 2 marks · Working and answer

Working and answer

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3. Question 2 · part c.i · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part c.i only.

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

3. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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4. Question 2 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

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DO NOT WRITE IN THIS AREA

2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

4. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

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5. Question 1 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.

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(c) **Figure 1** shows a network topology.

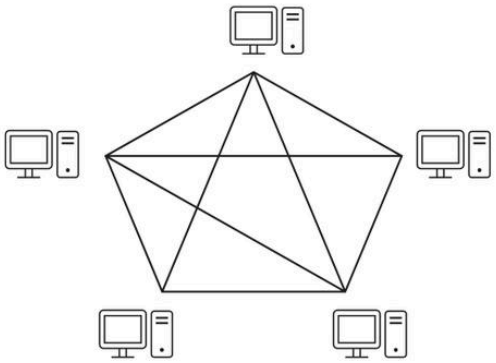


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

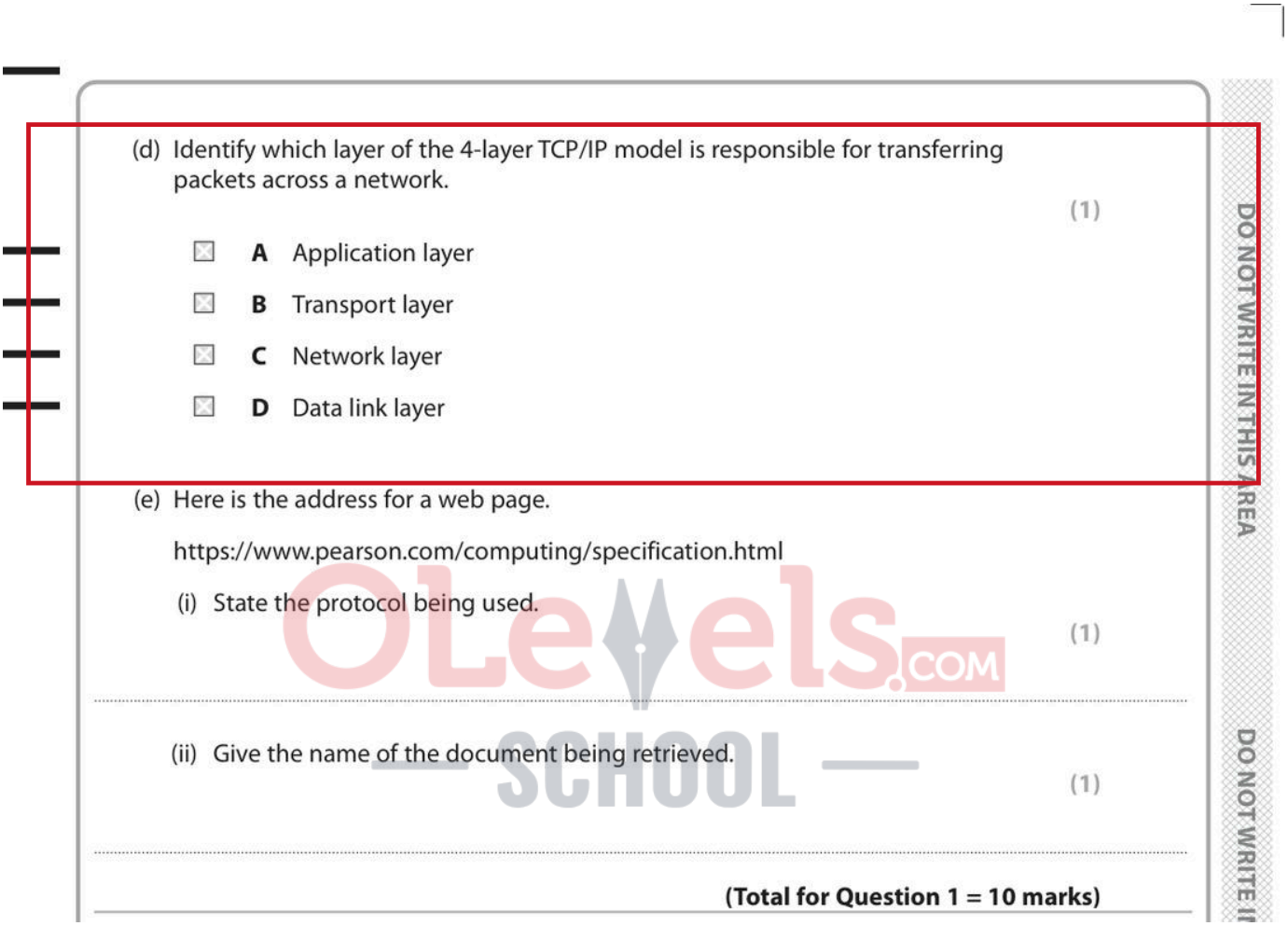
(2)

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5. Question 1 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.



(d) Identify which layer of the 4-layer TCP/IP model is responsible for transferring packets across a network.

(1)

- ☒ **A** Application layer
- ☒ **B** Transport layer
- ☒ **C** Network layer
- ☒ **D** Data link layer

(e) Here is the address for a web page.

https://www.pearson.com/computing/specification.html

(i) State the protocol being used.

(1)

(ii) Give the name of the document being retrieved.

(1)

(Total for Question 1 = 10 marks)

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Working and answer

6. Question 1 · part b · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

1 Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

(4)

Statement	DNS	PAN	WAN	WWW
A network of networks **not** limited to a single location	☒	☒	☒	☒
A network that connects devices within the range of a single user	☒	☒	☒	☒
A set of hypertext documents accessed through the HTTP protocol	☒	☒	☒	☒
A system that maps domain names to IP addresses	☒	☒	☒	☒

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

(2)

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

(1)

☒ **A** HTTP

☒ **B** IMAP

☒ **C** POP3

☒ **D** SMTP

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P 7 8 6 3 7 A 0 2 2 0

6. Question 1 · part b · 2 marks · Working and answer

Working and answer

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